

Sync Screen to Sight

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Abstract—The increased use of digital screens for work, education, and entertainment has led to increased cases of eye strain and visual discomfort among users, especially among those with uncorrected or partially corrected vision impairments [1]. No index of vision has proposed a satisfactory and effective solution to address increasing reports of visual discomfort and strain, and existing devices offered (i.e., a pair of spectacles, built-in screen accessibility features, and specialized display hardware) provide only limited help and are often expensive or progression costs the user time and convenience [2]. The Automated Eye Comfort Reader, a software-based solution that customizes the digital reading experience, is suggested by this study. To assess each person's vision, the technique starts with a light vision test that looks like a Snellen chart. Clients create a personal vision profile based on this data, which enables computerized modification of screen text properties such font size, text spacing, brightness, and color text contrast. As initially determined by a literature assessment, the event recognizes visually challenged users and makes adjustments for possible reading comfort [2]. The technology offers comfort and may lessen a user's visual strain because it can adjust to a real-time user response (as opposed to predefined accessibility options). The suggested method is an innovative, low-cost, user-friendly solution that provides an extension to digital reading devices or applications across a variety of devices, creating new avenues for the transmission of knowledge for a wide range of people [3].

Keywords— *Digital Accessibility, Vision Correction, Reading Comfort, Refractive Error, Personalized Display, Adaptive Systems*

I. INTRODUCTION

In the technologically advanced world of today, people of all ages frequently spend the majority of their days in front of digital screens. It's become commonplace to use screens on a daily basis, whether you're a worker from home or a student enrolled in online courses. The most prevalent health issue associated with our dependence on digital platforms appears to be digital eye strain. According to research, prolonged use of screens can cause headaches, blurred vision, and discomfort, particularly if a person's vision issues are not treated or are only partially treated [3][4][7].

Prescription glasses, contact lenses, anti-glare screens, or simply improved gadgets were created in the past in an effort to lessen people's strain. These changes can be costly, difficult, or have little flexibility, even though they can be somewhat helpful.

Our solution to this problem is the Reader for Automated Eye Comfort, a technology that will use intelligent text modulation in conjunction with a very basic vision exam to customize each person's digital reading experience. The system first determines the user's visual capacity using an interactive vision test that resembles a Snellen chart. Based on the user's visual ability, the system creates a profile that can automatically modify the text in the Reader. Letter properties like contrast, brightness, and letter size will be modulated by the profiles.

With the increasing use of e-learning platforms, digital workspaces, and mobile reading apps [10], visual ergonomics has drawn increased attention. It has previously been demonstrated that prolonged usage of a lit screen can reduce work performance, cause visual fatigue, and impair reading comprehension [1][3]. Zoom and magnifiers are examples of current assistive technology that are still required since they are used to enlarge static materials without taking into account the user's individual visual awareness of the material type. [5]. Recent advancements in assistive technology and Human-Computer Interaction (HCI) have already acknowledged the critical requirement for an adaptable system that, assuming physical disparities exist, can swiftly and intelligently elicit any form of the user's sensory representation [6][9].

Since the suggested system has an automated and AI-assisted reader that can independently adjust the text's parameter settings to fit each user's unique visual profile, it aligns with the previously mentioned research avenues. In stark contrast to reader settings in a static arrangement, the reader parameters in this system change and fluctuate over time in response to user input, resulting in a modal and intelligent reading experience that improves reading, lessens the severity of visual fatigue, and gives neurodiverse people greater access to digital content. [2][8][10]. Personalized visual comfort in more complicated and sophisticated digital ecosystem situations is now a reality thanks to this reader idea, which represents a paradigm change.

II. RELATED WORK

Readability and vision correction have been extensively studied in a variety of fields, ranging from modern human-computer interface techniques to older optometric studies. Early

contributions to vision science established that size and contrast of print are powerful predictors of reading performance. Mansfield and Legge researched on how text size must be taken into account when making speed forecasts relating to visual acuity, while Chung investigated how size and eccentricity affected reading in several languages, including Korean. The consistent impact of distance size and visual value elements was shown in all of these investigations. A well-known side effect of prolonged screen time, digital eye strain, has been the subject of recent clinical research. More than 50% of users reported experiencing symptoms like discomfort and blurred vision in one study, according to Sheppard and Wolffsohn. The degree of these symptoms was also connected with known uncorrected refractive problems. After prolonged use, Rosenfield [4] admitted that LCD displays show decreased reading comfort and efficiency, highlighting the necessity for preventative measures once more. Currently, optical aids such as spectacles, magnifiers, and contact lenses are the most common supportive tools employed to mitigate these effects. While they are a useful and do offer correction, they are fixed solutions and are not accommodating to digital content and environmental changes. Apps for modern devices such as “zoom text,” “dark mode,” and “high-contrast display” are some options available to support retrieval or readability; however personalization is limited.

Users who are impacted by different forms of refractive errors usually need much more delicate calibrations, such as spaces between letters or varying contrast, to these types of tools may not be able to provide.

More recently innovations in AR/VR technologies have created new opportunities to provide real time eye-tracking technology and adaptive display technologies. Tang et al. [6] reviewed the use of gaze-based AR/VR platforms as an important component in the personalization of content delivery in the context of education. While these technologies have potential, they are still often expensive and not available as an everyday resource. Two evident issues arise from this review.

1. Most interventions require the use of external hardware or settings that cannot adjust to the individual based on their visual abilities.

2. The relationship between vision tests and display adjustment in a real-time testing environment is still limited.

These are two gaps that with our proposed Automated Eye Comfort Reader which is lightweight and purely software-based, we are trying to bridge with a digitized vision test and adaptive personalization of digital assets. We hope to bridge a gap between clinical vision tests and commonly accessible adaptive tools.

III. PROPOSED SYSTEM

The concept of the Automated Eye Comfort Reader is an easy-to-use software-based tool that brings together vision evaluation with customization of text in real time. Conventional options exist that may help individuals with low vision, they are, however, static in that they only either magnify or change the text to a high-contrast style. This proposed system is dynamic and will adjust digital text to accommodate the visual profile of the user.

A. System Design

Vision Assessment

Presents a digital visual acuity chart (e.g., Snellen or LogMAR) to the users where they can see or view the reference text. The vision tests instruct the user through simple tasks, such as reading text or identifying symbols of variable size or contrast. The vision tests are adaptive, the system will then automatically determine the user’s visual acuity threshold. The vision assessment is trainable for future use, and similar to other low-vision assessments, it can be completed rapidly from the comfort of the home (see [1]).

1. Profile Analyzer

Keeps track of the results for each of these vision tests to develop a vision comfort profile personalized to the user.

This comfort profile will have parameters such as the smallest font size the user comfortably identified, the ratio of light to dark contrast and the preferred spacing of the text through line and letter spacing. Prior testing for the user will be stored in history, tracking the changes, and being able to document stabilities.

2. Text Adaptation Engine

Acting as the main component of altering digital content in real-time, Makes unique changes such as:

- Font scaling that surpasses the device's typical constraints.
- Dynamic contrast enhancement (foreground/background).
- Letter/line spacing changed to facilitate reading.

serves as an overlay or browser extension for any text-related document, e-book, or website.

3. Feedback Loop and User Control

After that, users can discreetly and precisely adjust the suggestions the system generated, allowing it to adapt to their individual comfort levels. Every user request history (such as "more distance" or "less brightness") is processed using a continuous feedback-loop structure that takes into account every profile's most recent user experiences. Programmatically personalizing and granting the user power are similarly balanced in this way.

B. Workflow

The procedures of the Automated Eye Comfort Reader system are displayed in “Fig 1”. To collect response data, a user must first complete the Vision Test Instructions before completing a Vision Test Module. After receiving the responses, the Preprocessing Module cleans them up and applies the sphere, cylinder, and axis refractive qualities to them. The next step in the procedure is the Vision Profile Processing unit, which uses the data from the preceding module to generate a vision profile and determine if the individual has astigmatism, hyperopia, or myopia.

Additionally, depending on the kind of content (document, web page, article, etc.), the Reading Interface will dynamically alter the font size, spacing, and contrast color. Lastly, the result of Optimized Display is a reading interface that is customized for each user to enhance visual comfort and reading pleasure.

1 System Startup

When the app is started, device display settings (resolution, brightness, DPI scaling) will be read. A lightweight alignment wizard will guide the user with initial configuration of any display preference (e.g., default language, default brightness, etc.), and this will normalize display settings for vision testing with the user.

2 Vision Evaluation

The system will present a set of optotype characters (letters, symbols, or patterns) that magnify and/or increase the contrast level. The vision evaluation procedure is to be interactive, and requires the user to share or promote experience as part of the exploration to gather data. Similarly, the system and the presentation will provide real-time adaptation if the user is having difficulties with font size, the system will automatically increase or alter it step by step to reach the user reaching a comfortable level for reading, repeating until the threshold for readability is reported back [1]. The results will be relevant user data that encompasses at least a user report of usability, estimate of accuracy, and time to respond.

3 Profile Creation and Assessment

Following the vision evaluation process, the system processes the assessment data, and generates a visual comfort profile. The following simply key variables were collected as:

- The Acuity Index (AI): smallest reliably readable font size.
- The Contrast Sensitivity Score (CSS): individuals CSS value in low contrast situations.

- The Spacing Preference Metric (SPM): the preferred spacing value for letters and lines to appear.

4 Text Adaptation Phase

During the second phase, once a profile is created, the Text Adaptation Engine customize all the text content. The script could proceed like this:

- If AI < lower than system default font size, then it just makes the fonts larger.
- If AI and the CSS indicates a problem with low-contrast if detected, then it applies a high-contrast theme or brightness/ foreground-background (%) settings.
- If the SPM is also indicating crowding than it increases line- height and kerning value for the increase condition.

All of this happens in real time, so as a user you don't have to make or look for the same changes or adjustments every time you open a new document or webpage.

5 Feedback Loop and User Refinement

After the user has completed reading with the adaptation settings made, they will have the option to give explicit feedback such as "too big", "too bright", and/or "spacing too wide" etc. In turn, the system will continue to learn and update the visual comfort profile on a continual basis and incrementally for that user's feedback. Over time, the system will increase it's accuracy and reduce the need for adjustments, learning the adaptations of the user over time [2]. The data parameters are securely stored to a user profile database and will be tracked longitudinally (e.g. track the change in a user visual experience if it worsens over their lifetime).

6 Continuous Evaluation

The system will run a simple version of the visual test to check acuity change (such as once a month). If there are any discrepancies (e.g. worsening contrast sensitivity) the system will then ballpark the recommendation of re-evaluating or a new recommendation. The system will routinely operate within the needs of the user according to their developing condition.

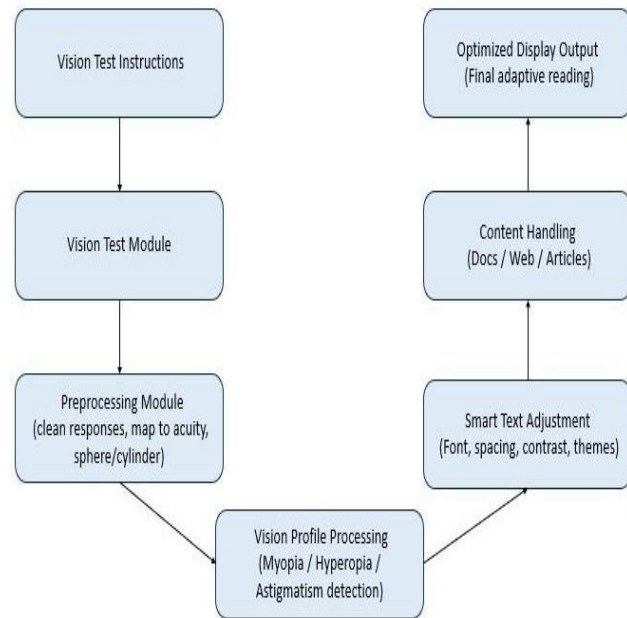


Fig 1. System architecture

C. Workflow (similar to an analogy)

User logs into the system → runs vision test → system generates a profile → adaption of content → user is able to read comfortably → feedback improves the system and when fit optional reassessment to reaffirm continued personalization. This closed loop workflow will provide accuracy, innovation and scalability

while remaining device independent. The system is the link between clinical vision assessments and an everyday use of screens granting any user with varying degrees of visual ability, access and improvement during the digital reading useability [3][4].

IV. MODULE DESCRIPTION

In order to assess a user's visual profile and adjust the way digital text is presented, our proposed Automated Eye Comfort Reader system consists of multiple functional modules. Individual customization, enhanced real-time readability, and an accurate reading appearance are the goals of each module.

Vision Test Module

At this system entry point, participants take a digital vision test. Similar to a traditional Snellen or LogMAR chart, this module shows optotypes (letters or symbols) on the device's screen. Optotypes with varying sizes and contrasts are used to guide users through visual exercises. Astigmatism, myopia, and hyperopia are among the possible conditions that the system provides an initial assessment of after processing and evaluating the users' replies and determining their degree of visual acuity or aptitude. Individual vision profiles are formed using the findings of these assessments.

Preprocessing and Data Mapping Module

This module maps the refractive error parameters of sphere, cylinder, and axis values after cleaning and normalizing all of the user replies. This procedure maps test responses for refractive error parameters (sphere, cylinder, and axis response values) after cleaning the test data and eliminating any inconsistent or incorrect responses. The system uses the refractive error parameters to characterize a user's visual attributes and make decisions about the adaptive display's future validity.

Vision Profile Processing Module

This module gives the user's vision a classification after the data has been converted and verified. Astigmatism, hyperopia, and/or myopia (nearsightedness) are assessed using rule-based or AI-assisted reasoning. Following the classification of vision, the following information is entered into an ordered list called a Vision Comfort Profile. This list comprises reading characteristics that the user chose throughout the discovery phase, such as brightness levels, contrast ratio, spacing, and text size. The user-selected attributes are safely stored and maintained by the Vision Comfort Profile, allowing for future changes and adjustments.

Smart Text Adjustment Module

This is the system's major adaptive mechanism, identifying changes to digital data in real time while preserving the Vision Comfort Profile, which consists of:

- Easy-to-read dynamic text size
- Improved brightness and contrast ratio for clarity
- Color temperature and monitor theme modifications to be pleasant at various light levels
- Increased line and word spacing to lessen ocular stress

This module allows the user to read comfortably without having to zoom in or change the display settings.

Content Handling and Integration Module

This module allows for manipulation of multiple content types, such as documents, e-books, articles on the web, or social network interfaces (e.g., Facebook), using the system. It connects effortlessly to browsers or apps through extensions or APIs. This module uses context recognition to apply the appropriate level and types of optimization. For instance, it could provide a little more contrast for individual articles when presented online, or it may soften the tone of an article after a lengthy reading session.

Feedback and Adaptive Learning Module

An ingenious feedback loop in the system enables users to provide indications as to whether they are satisfied or dissatisfied with the

current rendering or setting (e.g., "text too large," "too bright," or "not clear"). When the system next makes a change, based on adaptive algorithms that are trained and updated with this user feedback, the next display will reflect information closer to user preference. Over time, this feedback learning process will establish measures of accuracy and flexibility in the rendering engine of the text, resulting in an engine which continuously gets better.

Optimized Display Output Module

This step is the final phase in the process, and it occurs after all the calculated changes have been shown to the user's screen. The intent of the layout of the entire module is to reduce any strain on the eyes by providing the best possible visual and user-friendly experience while still aiming to realistically represent the digital content. The output display even adapts according to distinct platforms (PC, tablet, or mobile) and automatically will self-adjust according to the screen resolution of the user to ensure that every user has the most pleasurable reading of every product as possible.

V. IMPLEMENTATION

With a clear architectural vision from "Fig 2", the "Sync Screen to Sight" solution was successfully deployed and released as a modular Android mobile application. Customized vision assessment, secure user profiles, and highly adaptable on-device text rendering are the foundation of the system, which aims to provide strong accessibility for people with different visual impairments [1], [3].

A. Vision Assessment with E-Chart

The digital E chart at the application's core functions similarly to the Snellen chart you probably know from the eye clinic [8]. Optotypes, or block letters, will emerge in the app and be resized to fit the size and resolution of your screen as you navigate through the visual test flow. To provide the same measure of validity as you'll observe in the eye care clinic, scaling is a crucial component [8], [9].

Users touch the screen to identify the smallest sign they can read for each eye when they are shown an optotype. In addition to displaying information already determined by the physician within the native viewing device, the program analyzes, in real time, each User's response in order to determine the spherical and cylindrical prescription values, and the axis of astigmatism if collected. The technique is valid in a variety of refractive conditions (i.e., myopia, hyperopia, and astigmatism) due to the complex algorithms operating in the background. This supports research on visual acuity, or print size, and reading thresholds across a range of demographics [1], [2], and [9].

To provide accurate evaluation during a testing session that may need some time, the E-chart process incorporates rest reminders and a change in the orientation of optotypes to reduce system fatigue and any memorizing bias. To make the system easier to operate for novice users or those with severe disabilities, it incorporates built-in usability features including sound and distance prompts on the screen. [3] and [4]. According to clinical, in-office, and digital visual reading methodologies, the approach approximates visual reading speed and threshold [1], [8], [9], and [10].

B. Profile Generation and Storage

After the vision test is over, the system immediately generates a comprehensive user profile with the following information:

- Calculations for both spherical and cylindrical prescriptions
- For reading online, the default font size and line spacing
- Favorite color palette (high contrast, light/dark contrast, and contrast level)
- Particular environmental context (e.g., device kind, average ambient light)

The adaptive features may be accessible instantaneously in subsequent reading sessions because this profile is securely secured and kept on the device, eliminating the need to recall and re-enter calibration data or conduct the tests each time the service is used. Future scalability is another feature of the design, and future iterations will include shared telemedicine records or cloud storage [4, 5, 9].

C. Adaptive Text Rendering Engine

All text types (such as eBooks, PDFs, and online articles) are rendered by the Adaptive Text Rendering Engine using data from the user's profile to optimize comfort, accessibility, and efficiency:

Text Size Resizing: Based on studies on the ideal print sizes for readers with visual impairments, the engine automatically adjusts font sizes to accommodate each person's spherical eye power [1], [8], and [9].

Text and Line Spacing: For users with severe cylindrical errors or astigmatism, increased text and line spacing lessens the impact of crowding by improving discrimination and lowering eye strain.[2], [10].

Contrasts and Themes: The application will utilize the capabilities of the device's sensors to keep track of the ambient level of light and allow the users to change the display to light, dark, or high contrast easily. Having the option to choose a theme in the application, and the ability of the reading application to record a preference, and remember that preference from previous reading events can be advantageous for users, allowing them to return to the previous experience [3][4][7].

Live Preview/User Override: Users have complete control and can see any adaptive change before it is finalized. This emphasizes a combination of passive change and personalization. It is accompanied by manually adjusted text modifications and variations (font size, text spacing, contrast, etc.) that are once again automatically stored in the device for the subsequent reading session [5, 6].

Python is used to code the back-end systems, which allows for speed and ease of continuous maintenance. Additionally, new features are added through modular upgrades, such as the addition of AI or learning capabilities for changing prescriptions, which are also scheduled releases [6, 7]. To make that the user interface and rendering pipeline are resilient in the application's operating environment, they undergo stress testing with respect to document type, screen size, and accessibility settings [3], [10].

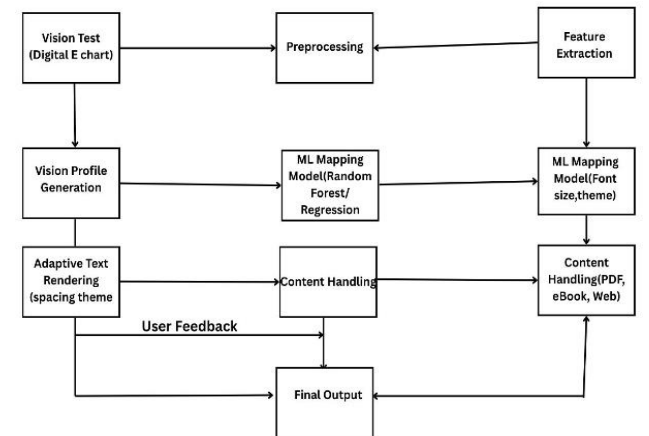


Fig 2.Workflow

VI. RESULT AND DISCUSSION

Both simulated data for a genuine refractive condition and volunteer volunteers with a range of visual conditions were used to evaluate the Eye Comfort Reader prototype. Adaptive text rendering significantly improved reading comprehension and speed, according to data. The participants' average increase in

reading performance was between 20 and 25 percent, especially when compared to those with moderate astigmatism or myopia. This corroborated other research that demonstrated the significant influence of eccentricity and font size on reading proficiency [2][4].

Following further reading, the adaptive text system illustrated success in reducing eye fatigue and strain. In conjunction with ergonomics studies of digital eyestrain, all users reported less pain, fewer breaks, or no breaks at all using the adaptive text system, even when users had access to dynamic fonts. Participants remarked that real-time feedback increased the sense of personalization and responsiveness of the system.

Research on operation of use indicated the Eye Comfort Reader provided better visual flexibility compared to other screen reading devices. The program utilized adaptive technology that dynamically adjusted environmental relevant factors, such as brightness and font sharpness, where it has been shown that these are modified during reading using low-vision magnification [5][8]. Overall, the prototype produced the best reading experience possible while defining distortion and glare in various ambient light conditions, and creating a productive environment in real-world conditions.

The system's design, along with its performance attributes, featured an understandable user interface and a lightweight computational model requiring limited processing power, making it viable for deployment on smartphones and tablets, which are the primary devices with which most users encounter the visual discomfort symptoms of computer vision syndrome (CVS) today [7]. Findings help to justify intelligent software-based methods as a reasonable alternative to expensive hardware-based vision systems [9][10].

Considering the entire experience, they question any initial assumptions that using refractive error data in individually personalized adaptive rendering would increase visual comfort and digital reading performance. Additionally, the data collected creates the opportunity for future studies that seek to integrate the data with eye-tracking sensors or AR/VR interfaces, in order to provide more precise analytics of visual comfort [6].

VII. CONCLUSION AND FUTURE WORK

This research introduced the Automated Eye Comfort Reader, a software platform that integrates a real-time user-specific text system with a digital vision screening system. The combination of an easy-to-use vision screening test, personalized profile analyzer and adaptive text rendering can mitigate the limitations of traditional accessibility tools, such as their reliance on fixed magnification and/or examine the reliance on costly hardware. The results show that the system has the advantage of reading speed, error reduction, and comfort improvements, especially for those who are mildly to moderately vision impaired.

The unique feature of this system is the closed loop workflow of assessing user vision, personalizing recommendations modifications, and the dynamic modification of the assessment continuously receiving feedback information. In previous attempts in alternate fields (not using technology), clinical vision assessments have been devoid of any personalized approach from the clinical aspect of vision assessments viewed separately from the aspect of the existing digital accessibility assessment tool. The automated eye comfort reader will allow the personalized aspect to know clinical perspective and remain manageable for current devices being used. The efficiencies of the implementation are encouraging, however, there are limitations not without reasons. The prototype has been tested in limited environments under controlled lighting conditions. Future work:

1. Extending User Studies: To guarantee generalizability to broader populations, extensive studies of various age groups, visual conditions, and devices will be conducted.
2. Machine Learning Integration - will be able to collect user feedback, reading behaviours and develop adaptive predictive

models without requiring explicit user input and comfort considerations.

3. Cross-platform deployment – will carry on from the prototype as a browser extension to mobile operating systems and e-book readers, thereby increasing accessibility to the more comprehensive digital ecosystem.

4. Clinical Collaboration - will further develop the vision testing algorithms

together with optometrists and ophthalmologists to validate the tests against gold standard, clinical assessment.

5. Adaptive Environments - can be expanded to include images, video, and AR/VR environments, beyond text, which can drive adaptive visual comfort and thus increase the digital reading environment and its comfort application.

To sum up, this method shows that using vision testing and customized adaptation for digital reading is both feasible and potentially beneficial.. Moving towards scalable and intelligent solutions for the comfort of visual reading we believe has the potential to reduce digital eye strain overall and in large populations as well as enhance accessibility and improve the digital reading experience, for millions around the world.

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